

A FIXED-RANGE OBSTRUCTION FOR QUASI MINIMUM-ATTAINING OPERATORS

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STATUS

This is a candidate full answer to the open question in Section 2, page 9, of Garcia–Maestre–Martin–Roldan [1]. The answer is positive: for every infinite-dimensional Banach range space Y , there is a Banach space X such that

$$\text{QMA}(X, Y) \neq \mathcal{L}(X, Y).$$

In fact, X can be chosen to be an equivalent renorming of the same underlying space Y .

SOURCE QUESTION

The source paper recalls that, for every infinite-dimensional domain space X , one can find a range space Y with $\text{QMA}(X, Y) \neq \mathcal{L}(X, Y)$, and then asks for the reversed, fixed-range form:

Is it true that for every infinite-dimensional Banach space Y ,
there is a Banach space X such that $\text{QMA}(X, Y) \neq \mathcal{L}(X, Y)$?

In [9, Theorem 4.17] it was shown that if X is any infinite-dimensional Banach space, then there always exists a Banach space Y such that $\text{QMA}(X, Y) \neq \mathcal{L}(X, Y)$. This was achieved by considering an infinite-dimensional separable subspace X_0 of X and setting $Y = X \oplus_1 (\ell_\infty \oplus_1 X/X_0)$. To finish this section we will show that the claim from that theorem can also be always achieved with some range space Y that is isomorphic to the original space X . We do not know, however, if for every infinite-dimensional space Y , there is always a Banach space X such that $\text{QMA}(X, Y) \neq \mathcal{L}(X, Y)$.

DEFINITIONS

For $T \in \mathcal{L}(X, Y)$, its minimum norm is

$$m(T) = \inf\{\|Tx\| : x \in S_X\}.$$

The operator T is quasi minimum-attaining if there are $(x_k) \subset S_X$ and $y \in m(T)S_Y$ such that $Tx_k \rightarrow y$. The class of such operators is denoted by $\text{QMA}(X, Y)$.

PROOF INTUITION

The obstruction is the inverse of a non-norm-attaining identity map. The Josefson–Nissenzweig theorem gives a weak-star null sequence $(y_n^*) \subset S_{Y^*}$. Weighting these functionals by scalars $t_n \uparrow 1$ defines an equivalent norm

$$\|y\| = \max \left\{ a\|y\|, \sup_n t_n |y_n^*(y)| \right\}, \quad 0 < a < 1.$$

The identity from the original Y into $(Y, \|\cdot\|)$ has norm one, because the weights approach one, but it cannot attain that norm: for each fixed y , weak-star nullity forces $y_n^*(y) \rightarrow 0$, while every finite weight is strictly below one. Inverting this identity gives an isomorphism from the renormed space into the original Y . If that isomorphism were QMA, the closed-range convergence would make the inverse attain its norm, contradicting the construction.

THE RESULT

Theorem 1. *Let Y be an infinite-dimensional real or complex Banach space. Then there is an equivalent norm $\|\cdot\|$ on the vector space Y such that, for $X = (Y, \|\cdot\|)$, the identity operator*

$$T = \text{Id} : X \longrightarrow Y$$

is bounded below but does not belong to $\text{QMA}(X, Y)$. Consequently, the inclusion $\text{QMA}(X, Y) \subset \mathcal{L}(X, Y)$ is proper.

Proof. By the Josefson–Nissenzweig theorem, there is a weak-star null sequence $(y_n^*)_{n \geq 1} \subset S_{Y^*}$. Fix $0 < a < 1$, and put $t_n = n/(n+1)$. Define, for $y \in Y$,

$$\|y\| = \max \left\{ a\|y\|, \sup_{n \geq 1} t_n |y_n^*(y)| \right\}.$$

This is an equivalent norm on Y , since

$$a\|y\| \leq \|y\| \leq \|y\| \quad (y \in Y).$$

Let

$$J = \text{Id} : Y \longrightarrow (Y, \|\cdot\|)$$

denote the identity map from the original normed space into the renormed space. We first show that $\|J\| = 1$ and that J does not attain its norm.

If $\|y\| \leq 1$, then $\|y\| \leq 1$, so $\|J\| \leq 1$. Conversely, given n and $\delta > 0$, since $\|y_n^*\| = 1$, there is $y \in S_Y$ such that $|y_n^*(y)| > 1 - \delta$. Hence

$$\|Jy\| = \|y\| \geq t_n(1 - \delta).$$

Letting $n \rightarrow \infty$ and $\delta \downarrow 0$ gives $\|J\| = 1$.

Now fix $y \in B_Y$. Since $y_n^*(y) \rightarrow 0$, choose $N \geq 1$ such that $|y_n^*(y)| \leq 1/2$ for all $n > N$. Then

$$\sup_{n > N} t_n |y_n^*(y)| \leq \frac{1}{2}, \quad \max_{1 \leq n \leq N} t_n |y_n^*(y)| \leq t_N < 1,$$

while $a\|y\| \leq a < 1$. Therefore

$$\|y\| \leq \max \left\{ a, t_N, \frac{1}{2} \right\} < 1.$$

Thus no point of B_Y attains the norm of J .

Set $X = (Y, \|\cdot\|)$, and let $T = \text{Id} : X \rightarrow Y$. Then T is an isomorphism and $T^{-1} = J$. Hence

$$m(T) = \frac{1}{\|T^{-1}\|} = 1.$$

The operator T cannot attain its minimum norm: if some $x \in S_X$ satisfied $\|Tx\| = m(T) = 1$, then $Tx \in S_Y$ and

$$\|J(Tx)\|_X = \|x\|_X = 1,$$

so J would attain its norm, impossible.

It remains only to exclude quasi minimum attainment. Suppose, toward a contradiction, that $T \in \text{QMA}(X, Y)$. Since $m(T) = 1$, there are $(x_k) \subset S_X$ and $y \in S_Y$ such that $Tx_k \rightarrow y$ in Y . Applying the continuous inverse $J = T^{-1}$, we get

$$x_k = J(Tx_k) \longrightarrow Jy \quad \text{in } X.$$

Therefore $\|Jy\|_X = \lim_k \|x_k\|_X = 1$, with $y \in S_Y$. This says that J attains its norm, the contradiction. Consequently $T \notin \text{QMA}(X, Y)$, proving the theorem. $\square$

VERIFICATION NOTES

The proof is analytic. The only external theorem used is the Josefson–Nissenzweig theorem, in the standard form that every infinite-dimensional Banach space Y admits a weak-star null sequence in S_{Y^*} .

The last paragraph also gives a direct replacement for the usual closed-range fact: for bounded-below operators, quasi minimum-attainment implies minimum-attainment.

NOVELTY CHECK

Local run indexes were searched for the arXiv id 2405.01302, the phrases “quasi minimum-attaining” and “minimum-attaining”, the expression “QMA of X, Y ”, and “Josefson–Nissenzweig”. No duplicate packet or attempt for this question was found. External web search on 2026-06-16 used the exact title of [1], the arXiv id, and phrases including “quasi minimum-attaining for every infinite-dimensional Y ”, “QMA(X, Y) minimum norm Banach Josefson”, and “minimum-attaining Josefson–Nissenzweig quasi”. The search found the source paper but no later paper explicitly answering this fixed-range question.

LIMITATIONS AND HUMAN REVIEW RECOMMENDATION

This solves the existential fixed-range question. It does not address the separate implication questions around Lindenstrauss properties A , A_M , and B_M mentioned earlier in the source paper.

For review, check the non-attainment argument for the renormed identity $J : Y \rightarrow X$, especially the use of weak-star nullity to make $\sup_n t_n |y_n^*(y)| < 1$ for each fixed $y \in B_Y$. Once that point is accepted, the QMA contradiction is immediate.

REFERENCES

- [1] D. Garcia, M. Maestre, M. Martin, and O. Roldan, *On density and Bishop-Phelps-Bollobas type properties for the minimum norm*, arXiv:2405.01302, 2024. <https://arxiv.org/abs/2405.01302>.
- [2] J. Diestel, *Sequences and Series in Banach Spaces*, Graduate Texts in Mathematics, vol. 92, Springer, 1984.